

NVH Source Locator — ■■■■■■■■■■ ■■■■

■■■■■-■■■■■

- ■ ■ ■ ■ ■ ■ ■ ■ ■ {#how-it-works}

NVH Source Locator ■■■■ ■■:

- **Coordinate system:** The coordinate system is used to define the position of the object in the 3D space. The origin is the point where the three axes intersect. The axes are labeled X, Y, and Z. The X-axis is the horizontal axis, the Y-axis is the vertical axis, and the Z-axis is the depth axis. The coordinate system is used to define the position of the object in the 3D space. The origin is the point where the three axes intersect. The axes are labeled X, Y, and Z. The X-axis is the horizontal axis, the Y-axis is the vertical axis, and the Z-axis is the depth axis.
 - **Units:** The units used to measure the position of the object are meters (m) for the X and Y axes, and meters (m) for the Z axis. The units used to measure the position of the object are meters (m) for the X and Y axes, and meters (m) for the Z axis.
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- 2 **Coordinate system** → 2D **Coordinate system** (X, Y)
 - 3 **Coordinate system** → 2D **Coordinate system** (X, Y)
 - 4 **Coordinate system** → 3D **Coordinate system** (X, Y, Z)

Coordinate system {#before-you-start}

Coordinate system:

- The coordinate system is used to define the position of the object in the 3D space. The origin is the point where the three axes intersect. The axes are labeled X, Y, and Z. The X-axis is the horizontal axis, the Y-axis is the vertical axis, and the Z-axis is the depth axis. The coordinate system is used to define the position of the object in the 3D space. The origin is the point where the three axes intersect. The axes are labeled X, Y, and Z. The X-axis is the horizontal axis, the Y-axis is the vertical axis, and the Z-axis is the depth axis.
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**NVH Source Locator**

TDOA Calculator

PRO



2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Pair A-B

Pair A-C

Step 1

Calibration — Speed of Sound

Sensor spacing A-B

-

118

+

cm

External knock time delay

-

471

+

μs

Calculated speed of sound

2505 m/s

Closest: PEEK (2500 m/s)

+ Save as custom material

Step 2

Event Measurement

Event time delay

⊕ trials

-

227

+

μs

Which sensor heard it first?

Sensor A

Calculate Source Location

Tab bar (scrolls horizontally on narrow screens)



NVH Source Locator

TDOA Calculator **PRO**

2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

Help

Free

partial

partial

partial

partial

partial

partial



	1D	2D
2-Sensor	2 1D	2-Sensor
3-Sensor	3 1D	
3-Sen+	3-Sen+ 1D	
4-Sensor	4 (A-B + C-D) 1D	
4-Sen+	4 2D, 4 1D, 4 LSQ	
3D	XYZ 4 1D	
3D+	6 3D, 6 LSQ	
Materials	Materials + 1D	
Help	Help 1D	










**XXXXXXXXXX XXXXX XXXXXX XXXXXX XXXX X XXXXX XXXXXX XX XXXXXXXX XX XXXXXXXX XX XXXXXX
XXXX (XX XXXX XXXXX)X**

2-Sensor ■■■ {#2-sensor-mode}

3-Sensor ■■■

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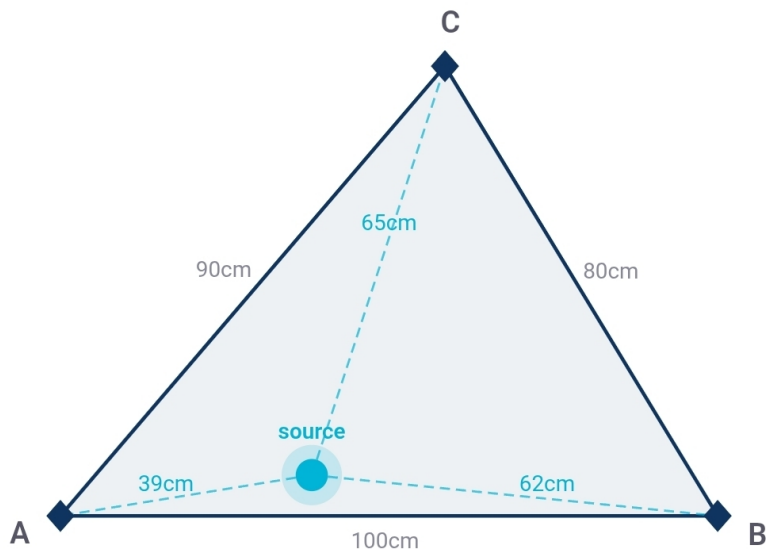
XXXXXXXXXX XXXXXXXX XXXX XXXXXX XXXXXXXX XXX XXXX XXXXXXXX XXXXXXXX XXXXXXXX, XXXXXXX, XXX
XXXXXXXXXXXX — XXX XXXXX XXXXXXXXXXXXXXXXXXXX XXX XXXXXXXXXXXX XXXX

■■■■ ■■■■ ■■■■

■■■■■■■■ ■■■■■■■■ ■■ ■■■■■■■■ ■■■■■■■■ ■■■■, ■■■■ ■■■■ ■■■■■■■■ (A-B, A-C, B-C) ■■ ■■■■ ■■■■■■■■ ■■■■ ■■■■■■■■

- **□□□□ □□□□□□: □□□□□□ □□□□ □□□□**

1. **Definition of the Problem:** A set of variables X, Y is defined over a domain D . The variables are represented by a vector $\mathbf{x}$ of size n . The domain D is a finite set of values. The variables are represented by a vector $\mathbf{x}$ of size n . The domain D is a finite set of values. The variables are represented by a vector $\mathbf{x}$ of size n . The domain D is a finite set of values.



 Annotate photo

THE UNITED STATES OF AMERICA - DISTRICT OF COLUMBIA

3-Sen+ (Pro)

3-Sensor 通过 3 个传感器 (A-B, A-C, B-C) 测量 3 个 TDOA 值。通过 3 个 TDOA 值可以计算出目标的 2D 位置 (X, Y)。

4-Sensor

4-Sensor 通过 4 个传感器 (A-B, A-C, B-C, C-D) 测量 4 个 TDOA 值。通过 4 个 TDOA 值可以计算出目标的 2D 位置 (X, Y)。

4-Sen+ (2D)

4-Sen+ (2D) 通过 4 个传感器 (A-B, A-C, B-C, C-D) 测量 4 个 TDOA 值。通过 4 个 TDOA 值可以计算出目标的 2D 位置 (X, Y)。

3D

3D 通过 4 个传感器 (A-B, A-C, A-D) 测量 3 个 TDOA 值。通过 3 个 TDOA 值可以计算出目标的 3D 位置 (X, Y, Z)。

3D+ (Pro)

3D+ (Pro) 通过 6 个传感器 (A-F) 测量 6 个 TDOA 值。通过 6 个 TDOA 值可以计算出目标的 3D 位置 (X, Y, Z)。

Materials {#the-materials-tab}

20 °C 材料属性表

NVH

NVH Source Locator

TDOA Calculator

PRO

2-Sensor3-Sensor3-Sen+4-Sensor4-Sen+3D3D+Materials

Typical longitudinal wave speeds at ~20 °C (sources: Evident/Olympus NDT & Dakota Ultrasonics). Actual speeds vary with composition, microstructure, temperature, and grain orientation – always calibrate in-situ for best accuracy. Tap a material to apply its speed across every tab – calibration Δt is computed from each pair's distance. Pairs without a distance entered are skipped.

Q

Search...

Air

343 m/s

ref only

☆

Teflon (PTFE)

1,400 m/s

ref only

☆

Mercury

1,450 m/s

ref only

☆

Water

1,480 m/s

ref only

☆

Silicone rubber

1,485 m/s

ref only

☆

Neoprene

1,600 m/s

ref only

☆

Motor Oil (SAE 30)

1,740 m/s

ref only

☆

Rubber, Butyl

1,800 m/s

ref only

☆

Polyurethane

1,900 m/s

ref only

☆

Glycerine

1,920 m/s

ref only

☆

LDPE

2,080 m/s

ref only

☆

Lead

3,460 m/s

ref only

☆

NVH

NVH Source Locator

TDOA Calculator

PRO

Platinum3,300 m/sref only☆

Tin3,320 m/sref only☆

Uranium3,378 m/sref only☆

Bronze3,530 m/s☆

Silver3,607 m/sref only☆

Ice4,000 m/sref only☆

Concrete4,000 m/sref only☆

Zinc4,170 m/s☆

Brass4,430 m/s☆

Iron (cast)4,572 m/s☆

Zirconium4,650 m/sref only☆

Copper4,660 m/s☆

Tungsten5,180 m/s☆

Glass (crown)5,300 m/sref only☆

Monel5,359 m/sref only☆

Nickel5,630 m/s☆

Quartz5,740 m/sref only☆

Materials

~340 m/s (~13,000 m/s)

14 20 °C

-
- (1020)
- (304)
- ()
-
-
-
-
-

- 温度补偿
- 温度补偿
- 温度
- 温度
- 温度
- 温度

温度补偿功能在 20 °C 时校准。校准温度 (T_{ref}) 必须在 -40 °C 至 +200 °C 范围内。

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温度补偿

2-Sensor 温度补偿功能在 20 °C 时校准。校准温度 (T_{ref}) 必须在 -40 °C 至 +200 °C 范围内。

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温度

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温度补偿 {#temperature-compensation}

温度补偿功能在 20 °C 时校准。校准温度 (T_{ref}) 必须在 -40 °C 至 +200 °C 范围内。

温度补偿

温度补偿功能在 20 °C 时校准。校准温度 (T_{ref}) 必须在 -40 °C 至 +200 °C 范围内。

Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

−

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.

0 / 500

DATA

🕒 Show intro screen again

ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.

□ □ □ □ □ □ □ □ □ □ □ □

■■■■■ ■■■■ ■■ ■■ ■■■■■■ $\neq 20\text{ }^{\circ}\text{C}$

- **Materials**
- toast: " (6,284 m/s @ 60 °C) — N "

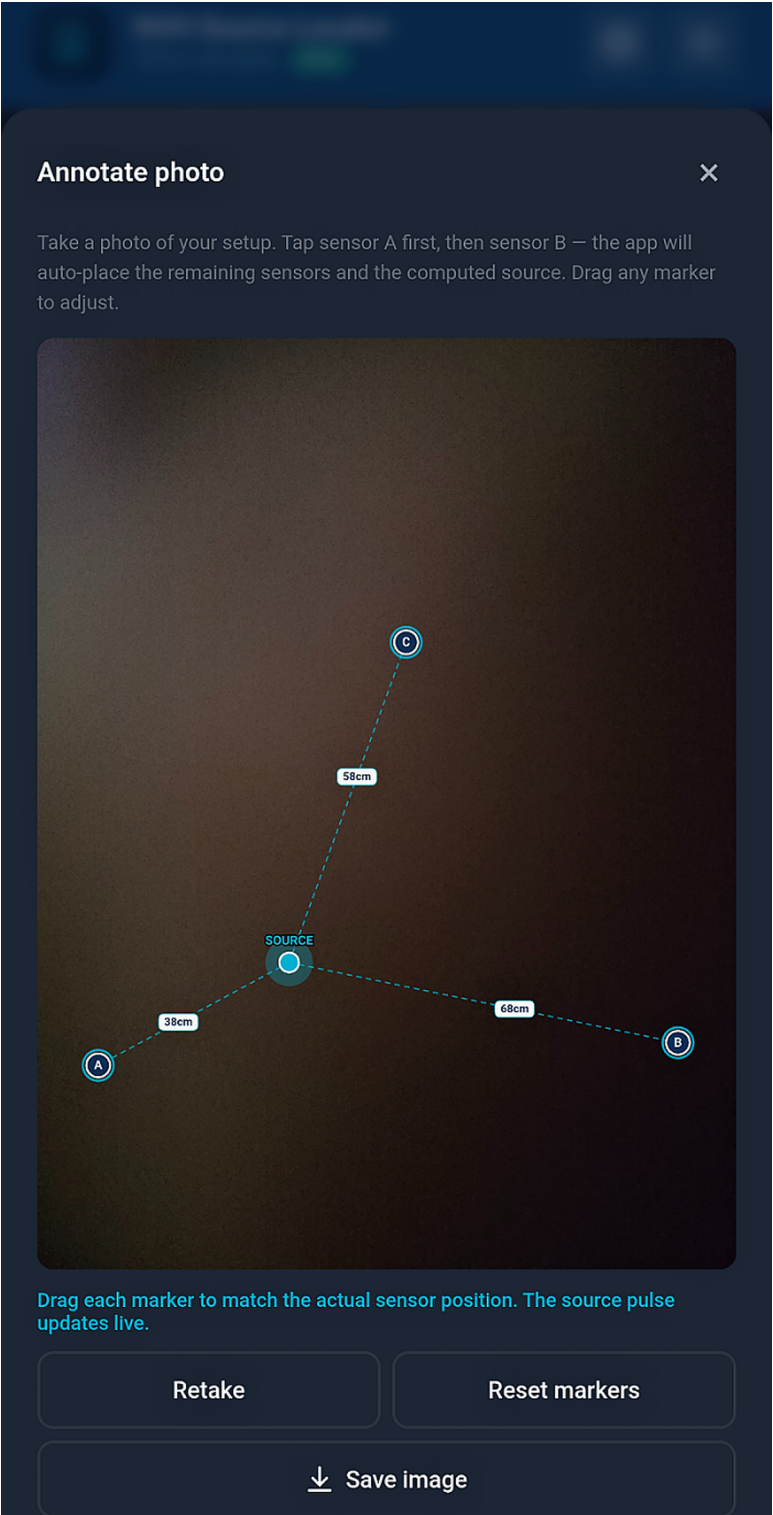
1. Die Temperatur der Luft beträgt 20 °C. Die Temperatur der Luft ist in der Tabelle angegeben.

THESE ARE THE TERMS AND CONDITIONS OF THE SALE OF THE GOODS TO YOU, AND YOU AGREE TO THESE TERMS AND CONDITIONS BY ACCEPTING THE GOODS. IF YOU DO NOT AGREE TO THESE TERMS AND CONDITIONS, YOU MUST RETURN THE GOODS TO THE SELLER WITHIN 14 DAYS OF RECEIVING THEM.

[illegible]

■■■■ ■■■■■■■■ {#photo-annotation}

THE UNITED STATES OF AMERICA, DISTRICT COURT OF THE DISTRICT OF COLUMBIA, hereby certifies that the foregoing is a true and correct copy of the original as the same appears on the records of the said Court.



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- ■■■■■■■■ ■■■■■■■■ (A, B, C, D, E, F ■■■■■ ■■ ■■■■ ■■■■ — 6 ■■■■■■■■ ■■) ■■ ■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■ ■■ ■■■■■■■■ ■■ ■■■■■■■■■■■■■■■■ ■■■■ ■■ ■■■■ ■■■■■ ■■■■

11 12 13 14 15 16 17 18 19 20 21 22 23

- **Android:** ■■■■■■ PDF ■■■■■■, ■■■■ ■■■■ ■■ ■■■■■■ ■■ ■■■■ ■■■■
- **iOS:** ■■■■■■ ■■■■■■ ■■■■■■ → PDF ■■ ■■■■ ■■■■ ■■■■■■, AirPrint, ■■ ■■■■ ■■■■

[illegible]

■■■■ ■■ ■■■■■■■■■■■■■■■■■■■■■■ {#backup-and-restore}

Page 10 of 10

[illegible]

11 12 13 14 15 16 17 18 19 20 21

XXXXXXXXXX → XXXXXXXXXXXXXXXX → XXXXX XXX XXX XXXXXXXXXXXXXX XXX XXXXXXX XXXXXXX XXXXXXX
XXX XXXXXXX XXXXXXXXXX, XXXXXXXXXX, XXXXXXXXXX XXX XXXXXXXXXXXXXXX XXXXX XXXXX XXXXX

THE INFORMATION CONTAINED HEREIN IS UNCLASSIFIED EXCEPT WHERE SHOWN OTHERWISE
 THIS DOCUMENT CONTAINS INFORMATION OF UNCLASSIFIED STATUS, AND SHOULD BE
 CLASSIFIED ACCORDING TO THE FOLLOWING INFORMATION: UNCLASSIFIED

■■■■■■■■■ {#settings}

Units & settings

×

LANGUAGE

English

DISTANCE

cm

inches

TIME

μs

ms

REFERENCE TEMPERATURE

Resets to 20 °C on next app launch

−

20

+

°C

Used for temperature-compensated material speeds. Always calibrate in-situ for best accuracy.

MEASUREMENT HISTORY

🕒 View recent calculations

⬇️ Backup

⬆️ Restore

REPORT

Report header text

e.g. EVDiag NVH Services · service@evdiag.net · +49-XXX-XXXXX

Appears at the top of every printed report.0 / 500

DATA

🕒 Show intro screen again

ABOUT

NVH Source Locator · TDOA Calculator for accelerometer-based source localization. Works offline. No data leaves your device.

Pro 	Pro (\$19.99)
 	(30)
 	, , ()
 	
 	, -40 °C +200 °C
 	



NVH Source Locator

TDOA Calculator



2-Sensor

3-Sensor

3-Sen+

4-Sensor

4-Sen+

3D

3D+

Materials

He

Place sensors A, B, C at the three corners of a triangle. Measure all three side lengths. Calibrate and measure pairs A-B and A-C. The app solves TDOA hyperbola intersection to locate the source.

▲ Show placement guide

Setup

Triangle side lengths

A-B side length



100

PRO



cm

A-C side length



90



cm

B-C side length



80



cm

Pro-■■■■■ ■■■■■

■■■■■

Questions? support@evdiag.net

THE UNITED STATES OF AMERICA, by and through the undersigned, its duly authorized representative, hereby certifies that the foregoing is a true and correct copy of the original document as submitted to the Commission on the part of the undersigned.

- PRO ■■■■ ■■ ■■■■ ■■ ■■■■
- ■■■■■■ ■■■■
- ■■■■■■ ■■ ■■■■ ■■■■■■ ■■■■ (\$19.99 ■■■■■■■■; ■■■■■■■■ ■■ ■■■■■■■■ ■■■■■■ ■■ ■■■■ ■■)

- **Android — iOS Apple** (Android — iOS Apple)
- **Android — iOS Apple**

Pro ■■■■■■

1. **Pro Version:** The Pro version of the app is available for purchase on the Google Play Store and the Apple App Store. It offers advanced features and a more comprehensive set of tools for creating and managing your content.

Pro

■■■■ ■■■■■ ■■ ■■■■■■■■ ■■ ■■■■■ ■■ ■■■■■ ■■ (■■■■ ■■■■) Pro ■■■■■ ■■■:

- **Google (Android)** **Apple ID (iOS)**
- **NVH Source Locator**
- **→**
- **Pro**

[Google Play Store](#)
[App Store](#)
[Source Locator](#)
[NVH](#)

[www.pearsoned.com/uk](#)
[www.pearsoned.com/uk](#)
[www.pearsoned.com/uk](#)
[www.pearsoned.com/uk](#)
[www.pearsoned.com/uk](#)

Android: ■■■■■■ ■■■■ "■■■■■ ■■■■ ■■■■ Google Play ■■■■■■ ■■■■ ■■?" ■■■■ ■■■■ ■■■■ ■■
■■■■ ■■■■■■-■■■ Google Play ■■■■■■■■■■ ■■■■ ■■ ■■■■■■ ■■■■

iOS: 3.1.1 Apple " " Google Play " "

Help {#help-tab-and-tutorials}

Help ■■■■ ■■■■ ■■-■■ ■■■■■■■■■■, ■■■■■■■■■■-■■■■■■ ■■■■ ■■ ■■■■■■ ■■■■■■■■
■■■■■■ ■■■■



NVH Source Locator

A mobile TDOA calculator that finds the exact origin of knocks, clicks and vibrations on any component – using accelerometers and an oscilloscope.

Works offline

Installable on Android & iOS

No data sent anywhere

WHAT EACH TAB DOES



2-Sensor

Single pair A-B or A-C. Linear result bar.



4-Sensor

Two pairs, proportional rectangle map.



3-Sensor

Triangle mode, TDOA hyperbola solver.



Materials

49 materials. Tap to auto-fill calibration.



Help

Mode explainers, placement guides, and a 10-item FAQ covering calibration, TDOA theory, and accuracy limits.

HOW IT WORKS

1

Calibrate — find the speed of sound

Strike the component outside the sensor pair. Enter the sensor spacing (cm) and the measured time delay (μs) — the app calculates the speed of sound in the material.

[Help](#) ■■■

■■■■■■■■ ■■ ■■■■ ■■■■ ■■■■:

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■■■■■■ ■■■■■■ {#troubleshooting}

Age Group	Percentage
18-24	10%
25-34	20%
35-44	15%
45-54	10%
55-64	10%
65-74	15%
75-84	15%
85+	15%

- **Точность измерения температуры** — tCal обеспечивает высокую точность измерения температуры — **±0,1°C**. Это позволяет точно контролировать температуру в процессе приготовления пищи.
- **Время приготовления** — tCal позволяет установить время приготовления пищи, что особенно полезно для приготовления блюд, требующих длительного времени.
- **Время ожидания** — tCal позволяет установить время ожидания, что особенно полезно для приготовления блюд, требующих длительного времени.

Toast "toast icon"

■■■■ ■■■■ ■■ ■■ ■■■■■ ■■■■ ■■■■■■ ■■ ■■■ ■■■■ ■■■ ■■■■■■■ ■■■■:

[illegible]

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 ■■ (50 m/s ■■ ■■ ■■ 20,000 m/s ■■ ■■■■■■) ■■■■■■ ■■■■■■ ■■ ■■■■■■ ■■■■■■ — ■■■■■■ tCal
 ■■ ■■■■■■ ■■■■ ■■■■■■■■

Materials

[illegible][illegible][illegible][illegible]

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